

Shaft Sizing: Combined Torsion + Bending

Shaft 1 — Input (from CBR primary reduction)		carries Stage 1 pinion
Torque at this shaft (from drivetrain)	63000 N·mm	
Estimated bending moment, M (assume $\cdot$)	31500 N·mm	
Combined shock factor, Kb (bending)	1.50	
Combined shock factor, Kt (torsion)	1.50	
Allowable shear stress, tau_allow (case-1)	153 MPa	0.75* 0.18 S _{ut} vs 0.30 S _y . FROM ASME2001
Minimum shaft diameter, d_min	15.21 mm	
Selected (rounded up) shaft diameter	20.0	Standard Shaft Diameter 15, 17, 20, 25, 30, 35
Shaft 2 — Intermediate		carries Stage 1 gear + Stage 2 pinion
Torque at this shaft (from drivetrain)	94563 N·mm	
Estimated bending moment, M (assume $\cdot$)	47282 N·mm	
Combined shock factor, Kb (bending)	1.50	
Combined shock factor, Kt (torsion)	1.50	
Allowable shear stress, tau_allow (case-1)	153 MPa	
Minimum shaft diameter, d_min	17.41 mm	
Selected (rounded up) shaft diameter	20.0	
Shaft 3 — Output (to differential)		carries Stage 2 gear
Torque at this shaft (from drivetrain)	141939 N·mm	
Estimated bending moment, M (assume $\cdot$)	70970 N·mm	
Combined shock factor, Kb (bending)	1.50	
Combined shock factor, Kt (torsion)	1.50	
Allowable shear stress, tau_allow (case-1)	153 MPa	
Minimum shaft diameter, d_min	19.94 mm	
Selected (rounded up) shaft diameter	25.0	